technische universität dortmund

Chair of Enterprise Computing
Department of Computer Science

Business Process Management

Process Technology

Christian Janiesch | Business Process Management | Dortmund 2023-07-03

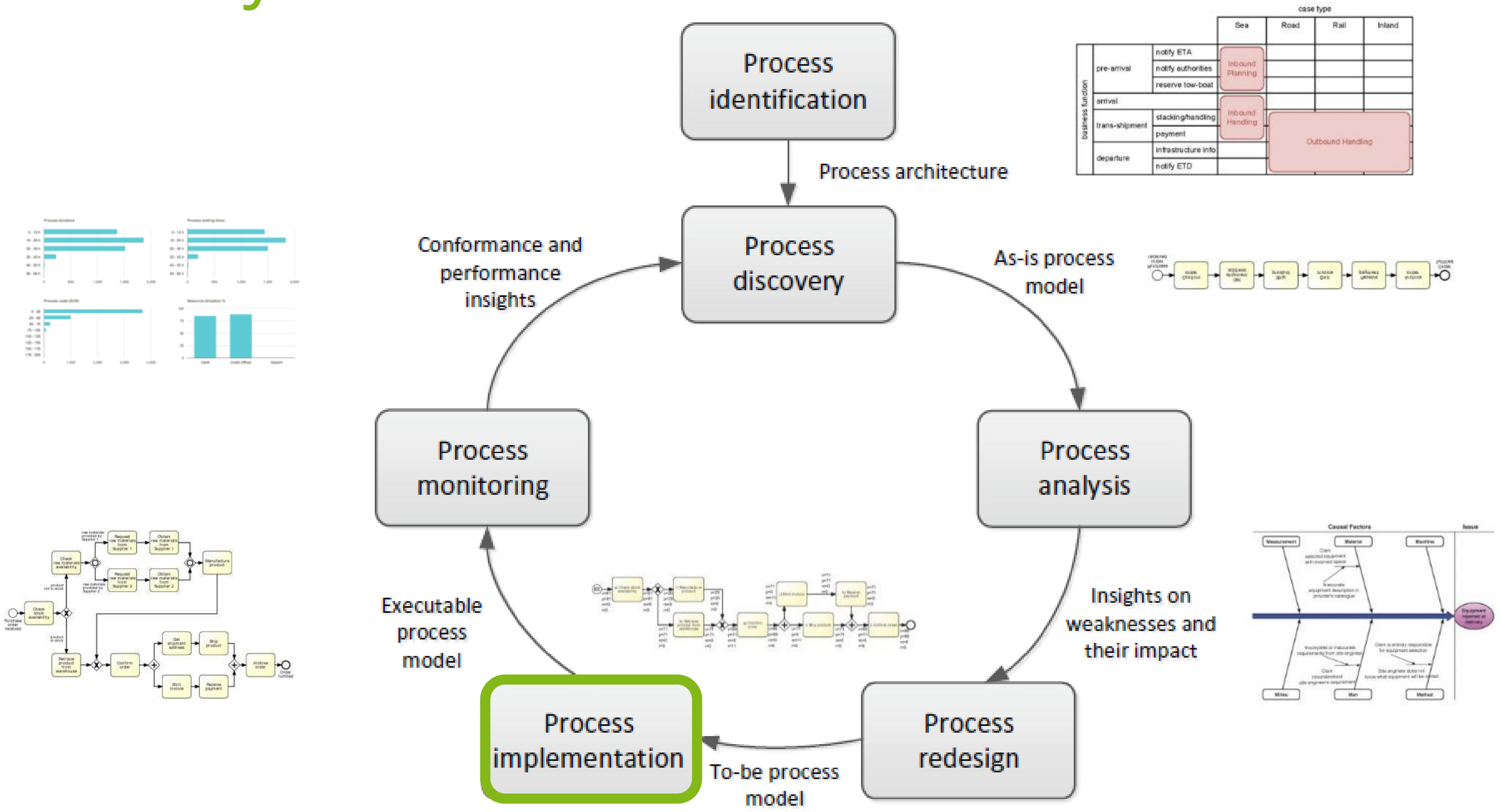
Roadmap

Introduction	10.04. Introduction	17.04. Process Identification			
Process Discovery & Modeling	24.04. Essentials of BPMN	01.05. Public Holiday	08.05. Advanced BPMN I	15.05. Advanced BPMN II	22.05. Process Discovery
Process Analysis & Redesign	29.05. Public Holiday	05.06. Qualitative Analysis	12.06. Quantitative Analysis	19.06. Public Holiday	26.06.2025 Process Redesign
Application of BPM	03.07. Process Technology	10.07. Process Monitoring and Mining	15.07. BPM as an Enterprise Capability		

Learning Goals of Today

- Understand the evolution of process technology
- Understand workflow execution semantics
- Understand BPMS
- Understand RPA

The BPM Lifecycle

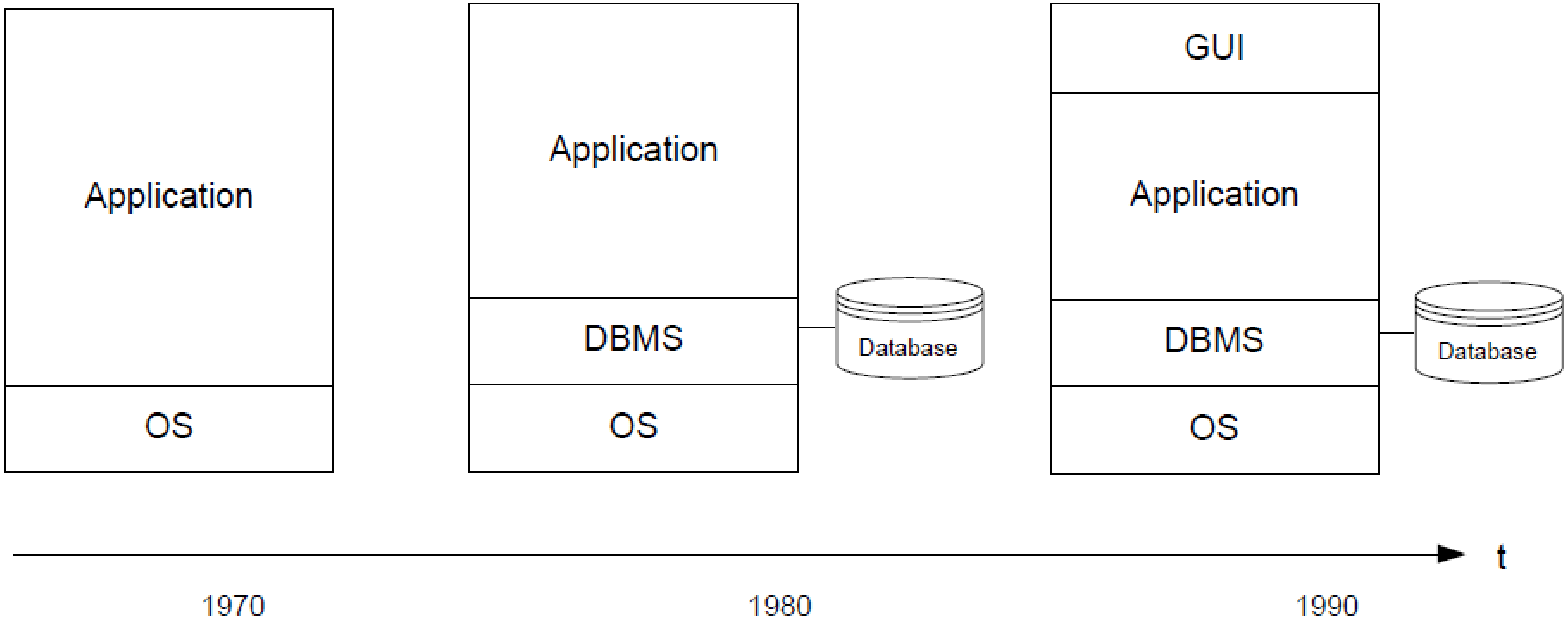


Dumas et al. (2018)

Evolution of Enterprise Software

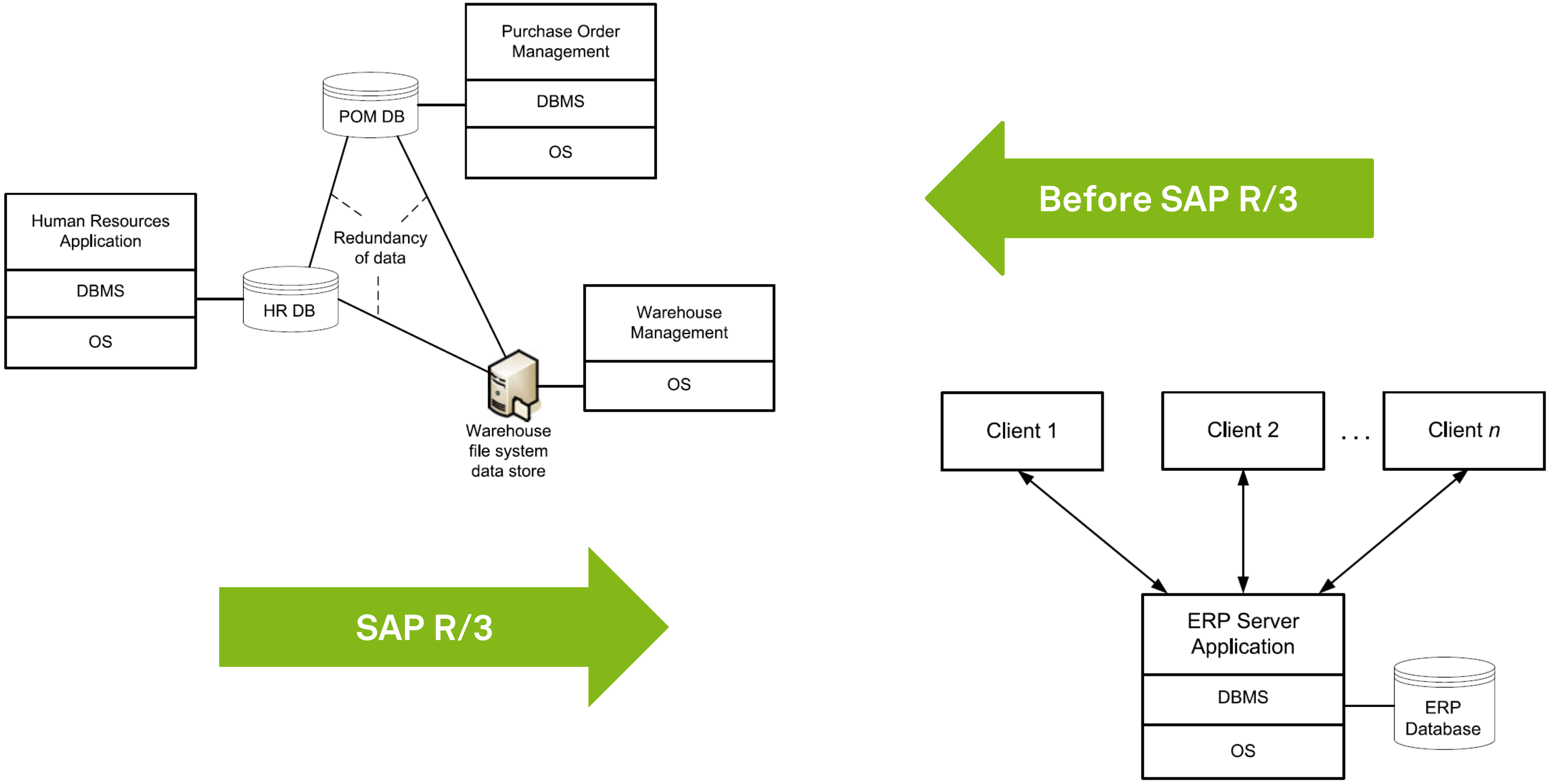
   

Software Applications Before BPM



Weske (2019)

Enterprise Software Before BPM



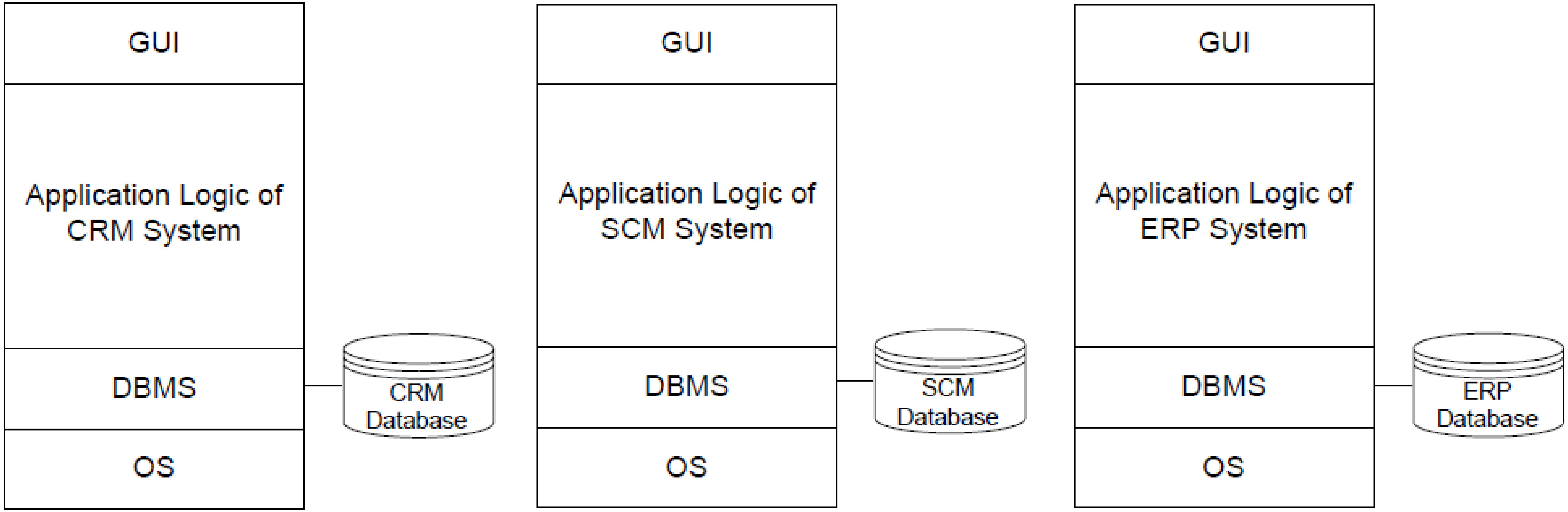
Weske (2019)

Domain-Specific Process-Aware Information Systems

- **Enterprise resource planning (ERP)** systems
  - Provide generic business functionality, which is required across various industries
    - accounting, controlling, human resource management, production, etc.
- **Customer relationship management (CRM)** systems
  - Support operational and analytical marketing and sales
    - sales and marketing activities related to pricing, distribution, and campaigning
- **Supply chain management (SCM)** systems
  - Focus on support of logistics operations
    - freight and transportation, warehousing, storage and inventory
- Product lifecycle management (PLM) systems
  - support the various processes of the lifecycle of a product

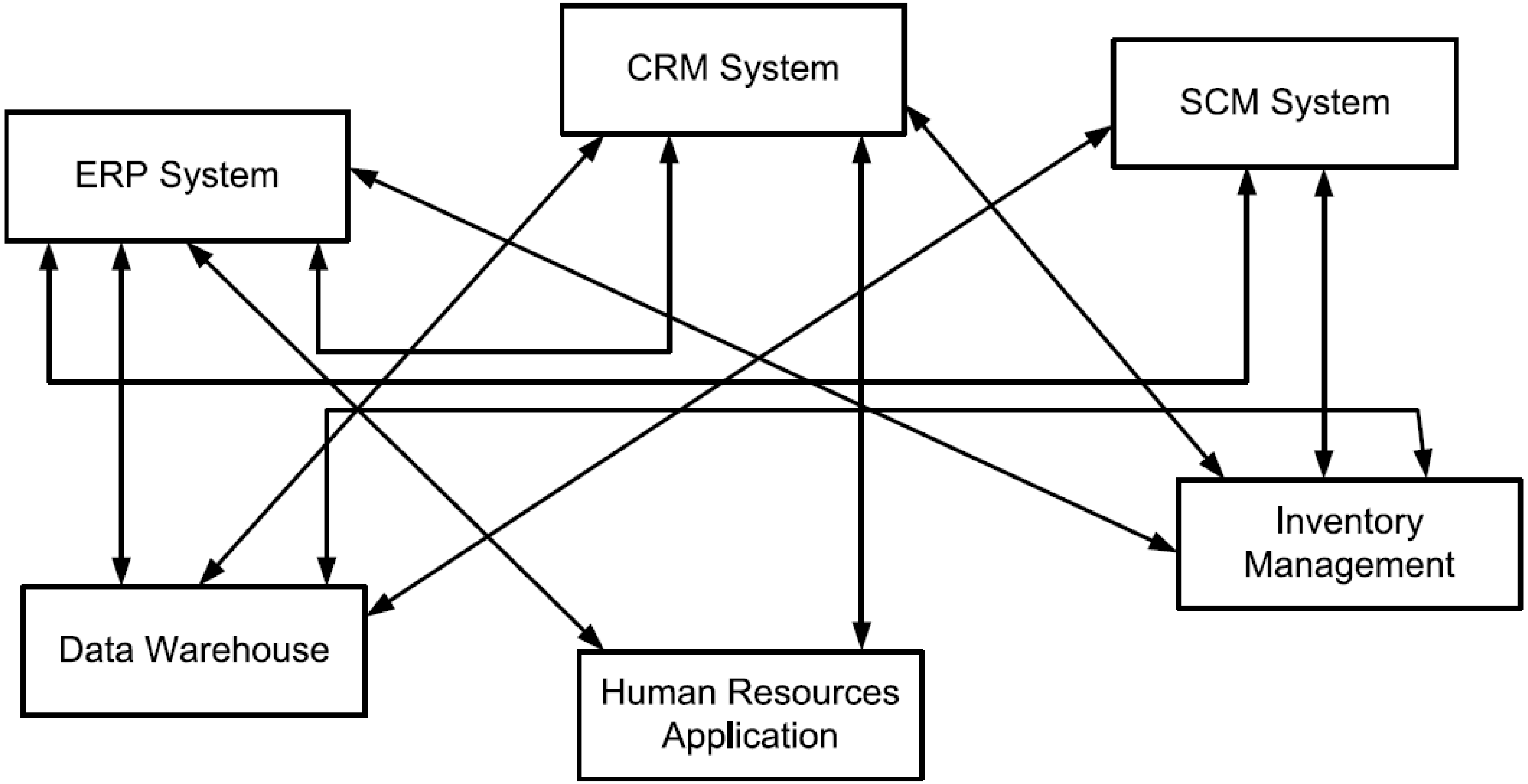
Dumas et al. (2018), Janiesch (2021)

Enterprise Software Before BPM



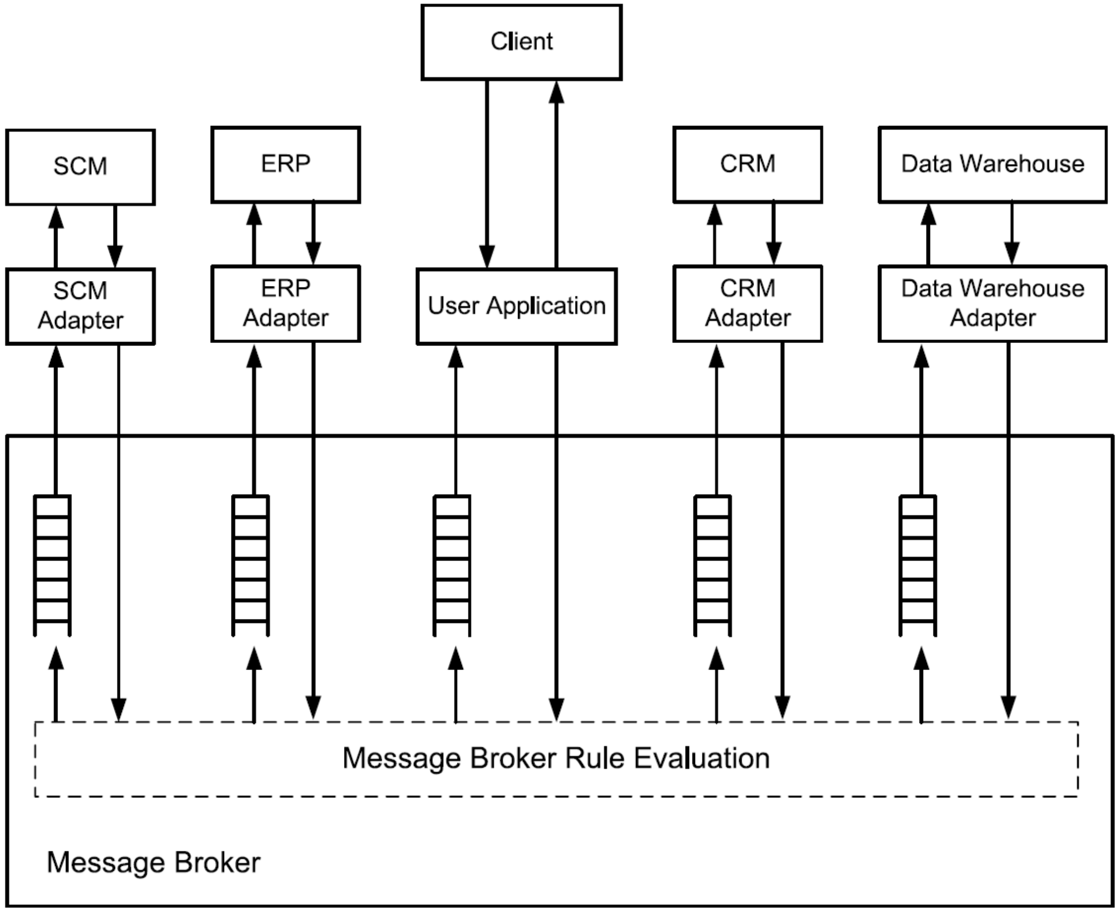
Weske (2019)

Enterprise Software Integration (1)



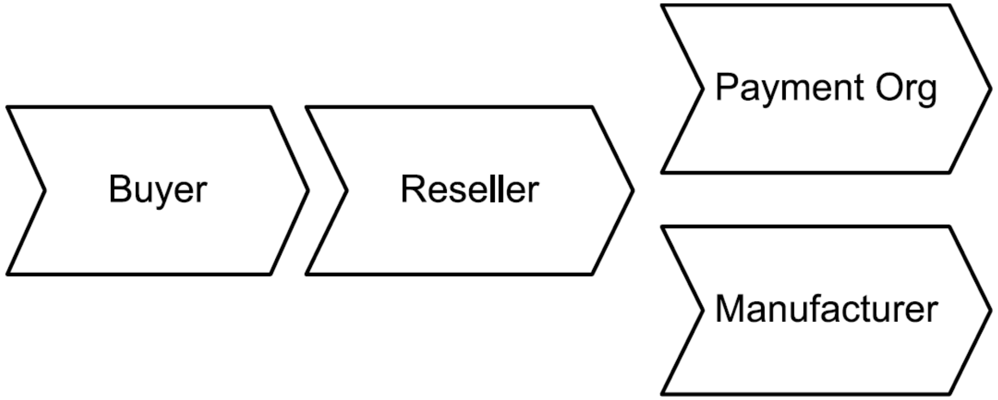
Weske (2019)

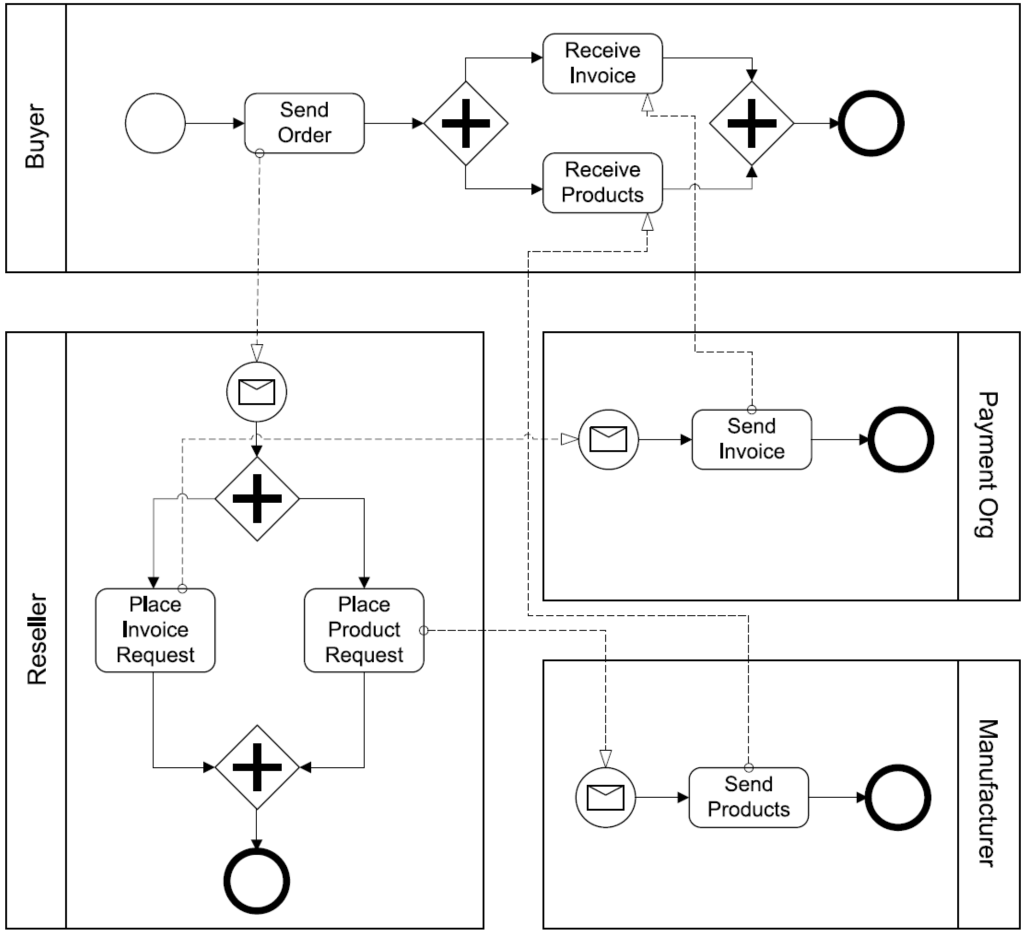
Enterprise Software Integration (2)



Weske (2019)

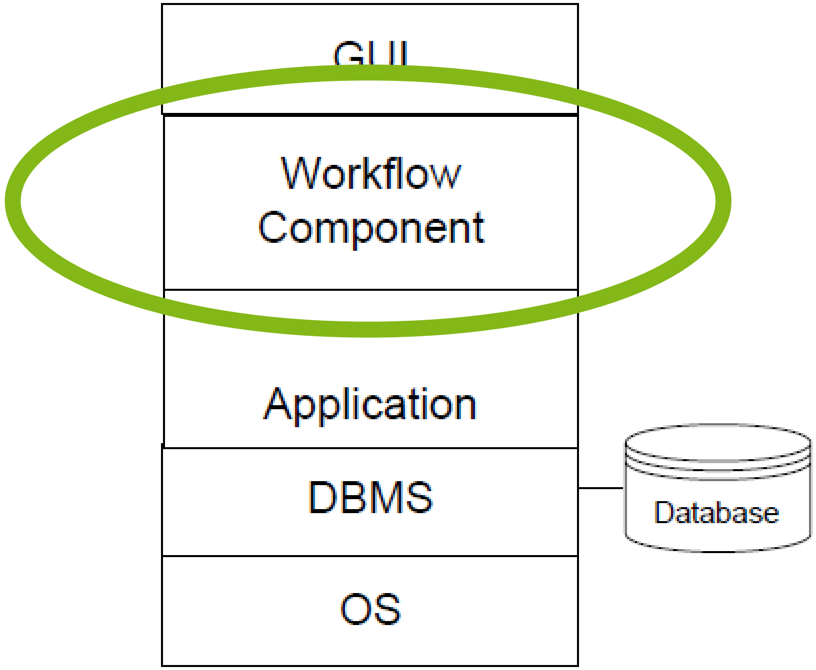
Value and Process Orientation





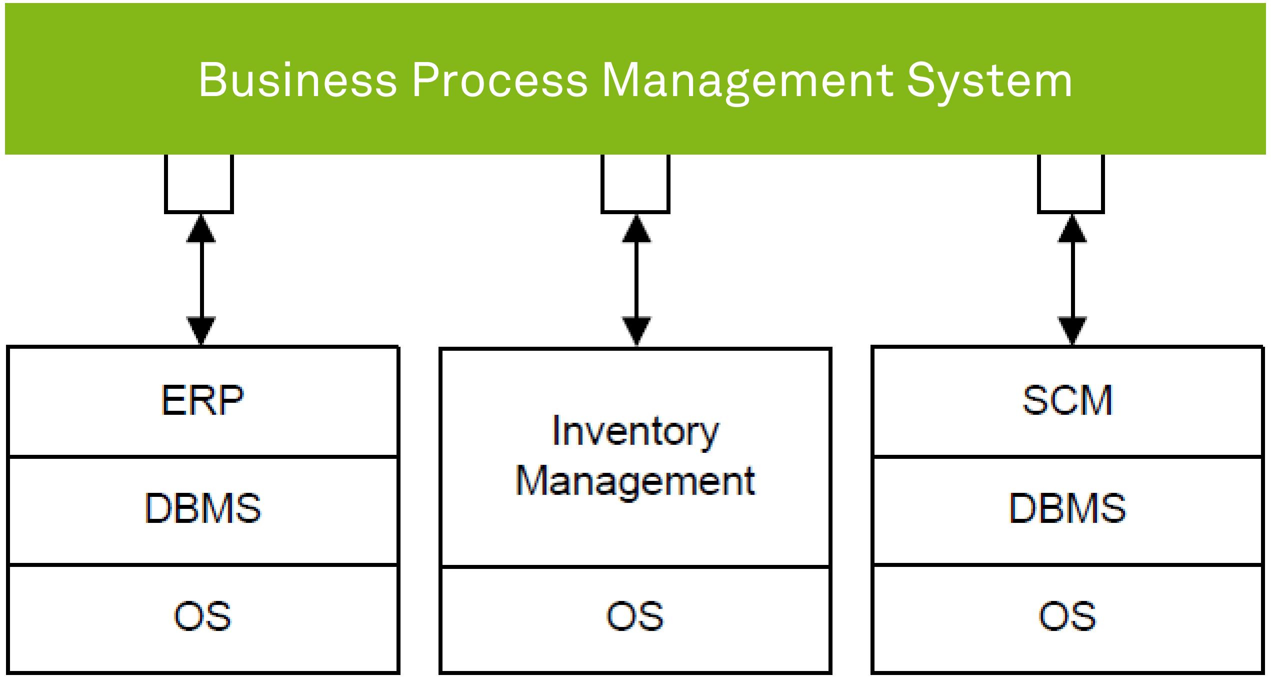
Weske (2019)

BPM/ Workflow: It's Another Layer!



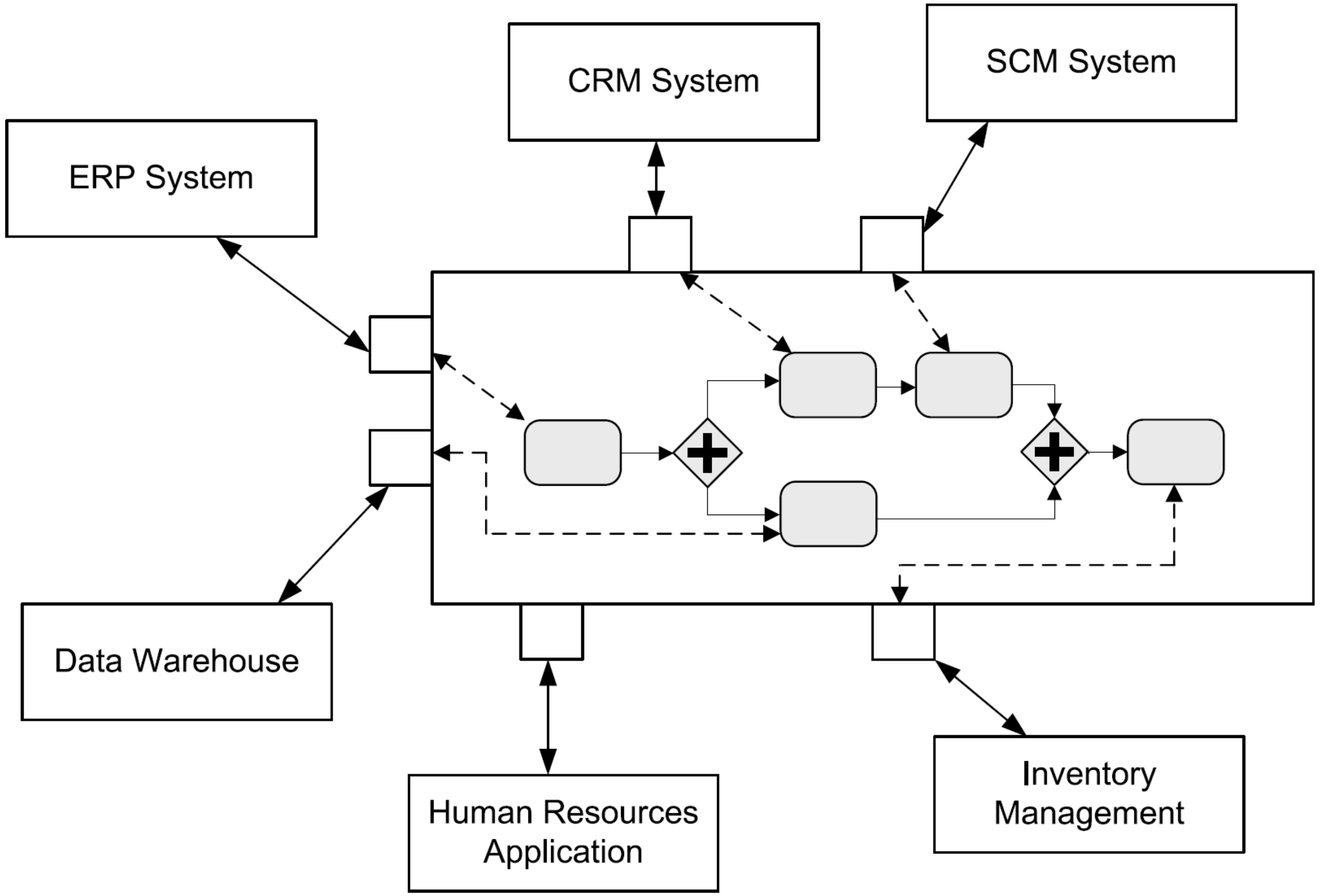
Weske (2019)

Process Orchestration with BPMS



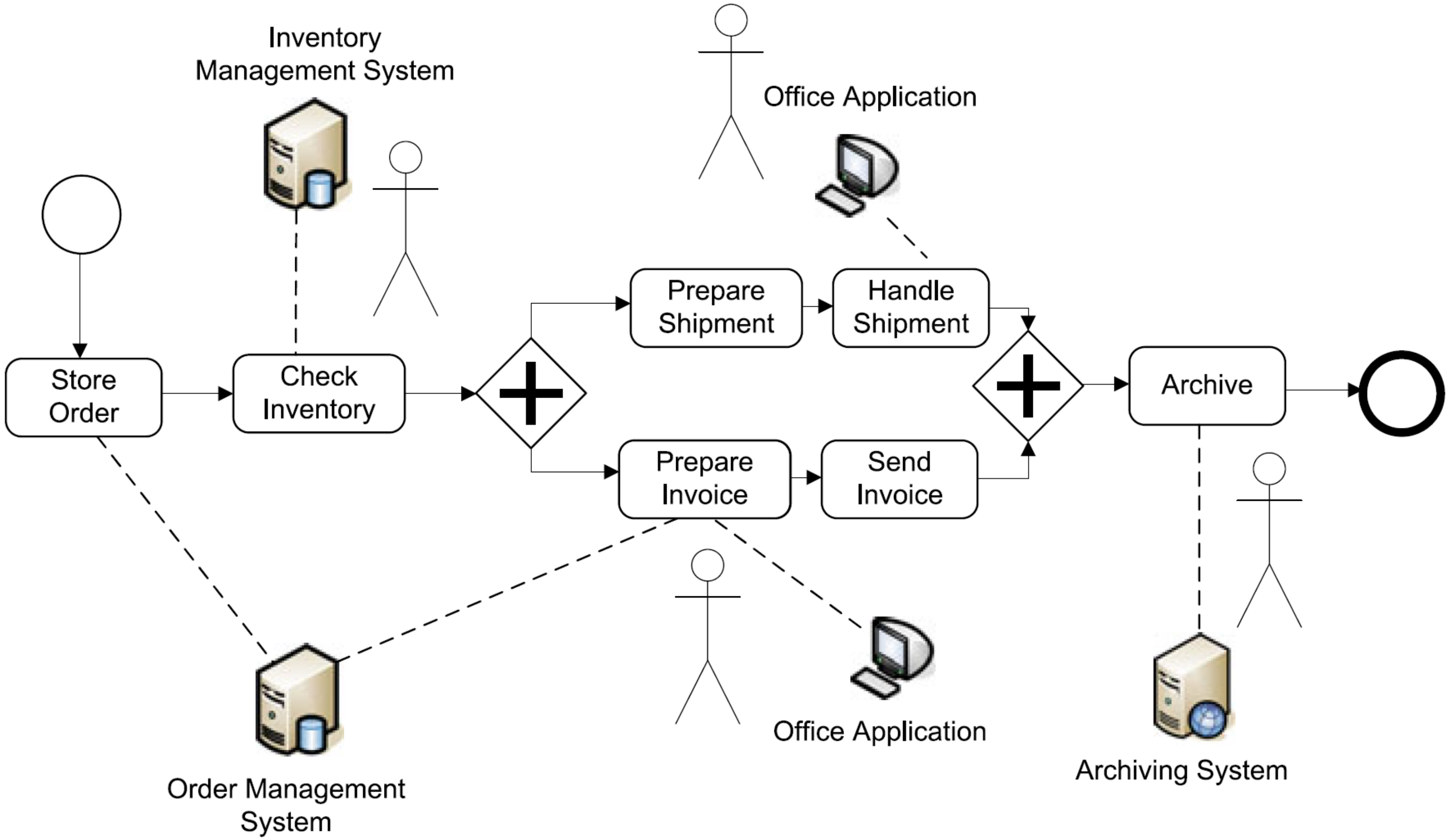
Weske (2019)

Process Orchestration with BPMS in Detail



Weske (2019)

Human Interaction



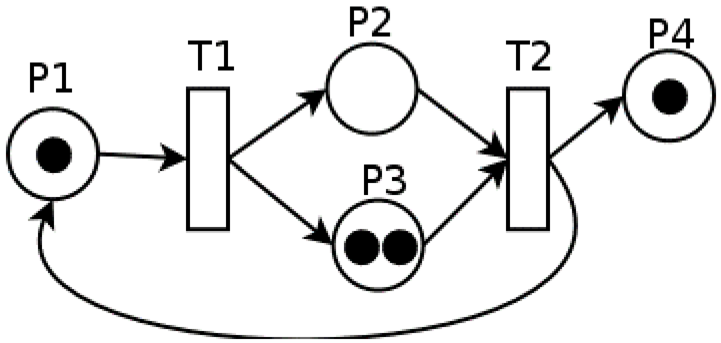
Weske (2019)

Execution Semantics




Execution Semantics

- Execution semantics are described by means of **token flows** (comparable to tokens in Petri-nets)
- Tokens flow along **sequence flow**
- Tokens are **produced and consumed** by activities, events and gateways



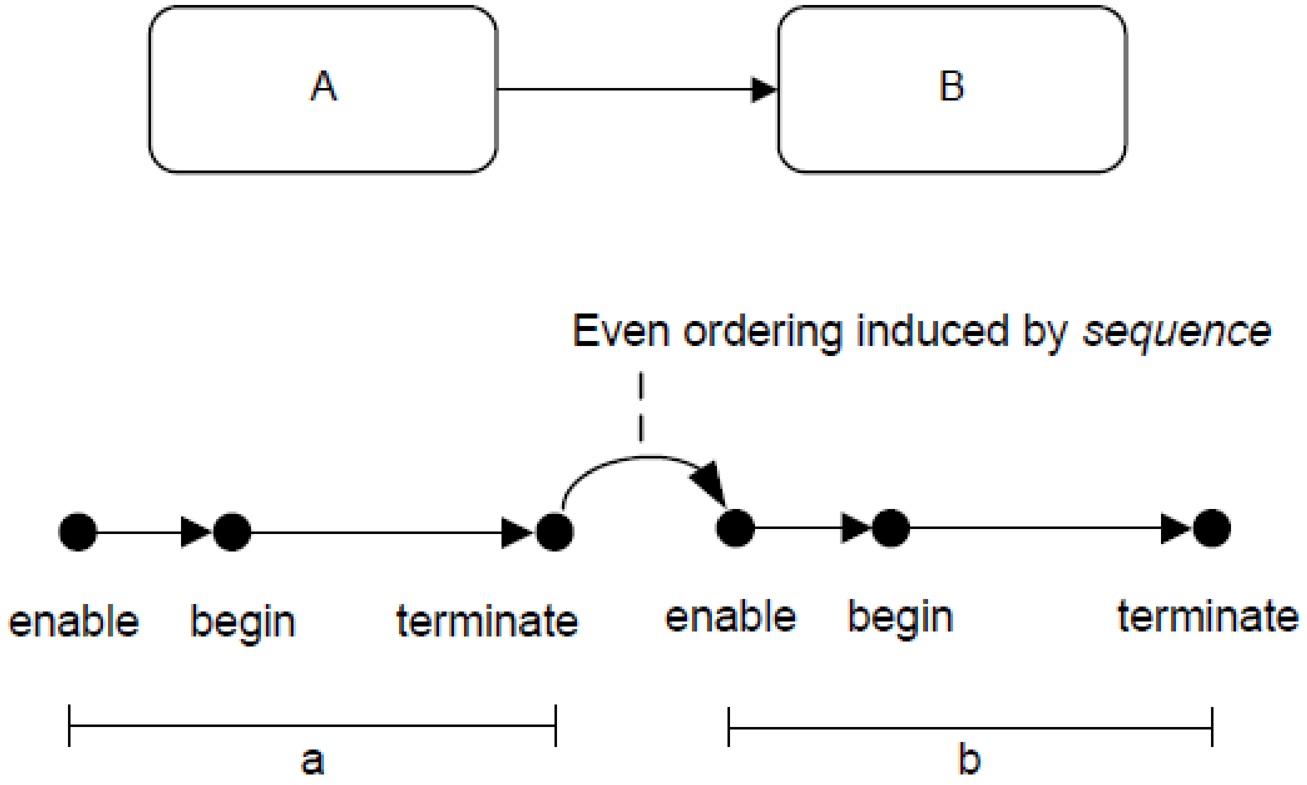
Weske (2019)

Process Lifecycle

- **Instantiation**
  - When one of its start events is triggered
  - Exception: If start event is part of an existing conversation that conversation is joined
- **Completed**
  - When all of the following hold
    - No remaining token
    - No active activity
    - If process was instantiated via parallel event-based gateway all events must have occurred

Object Management Group (2011)

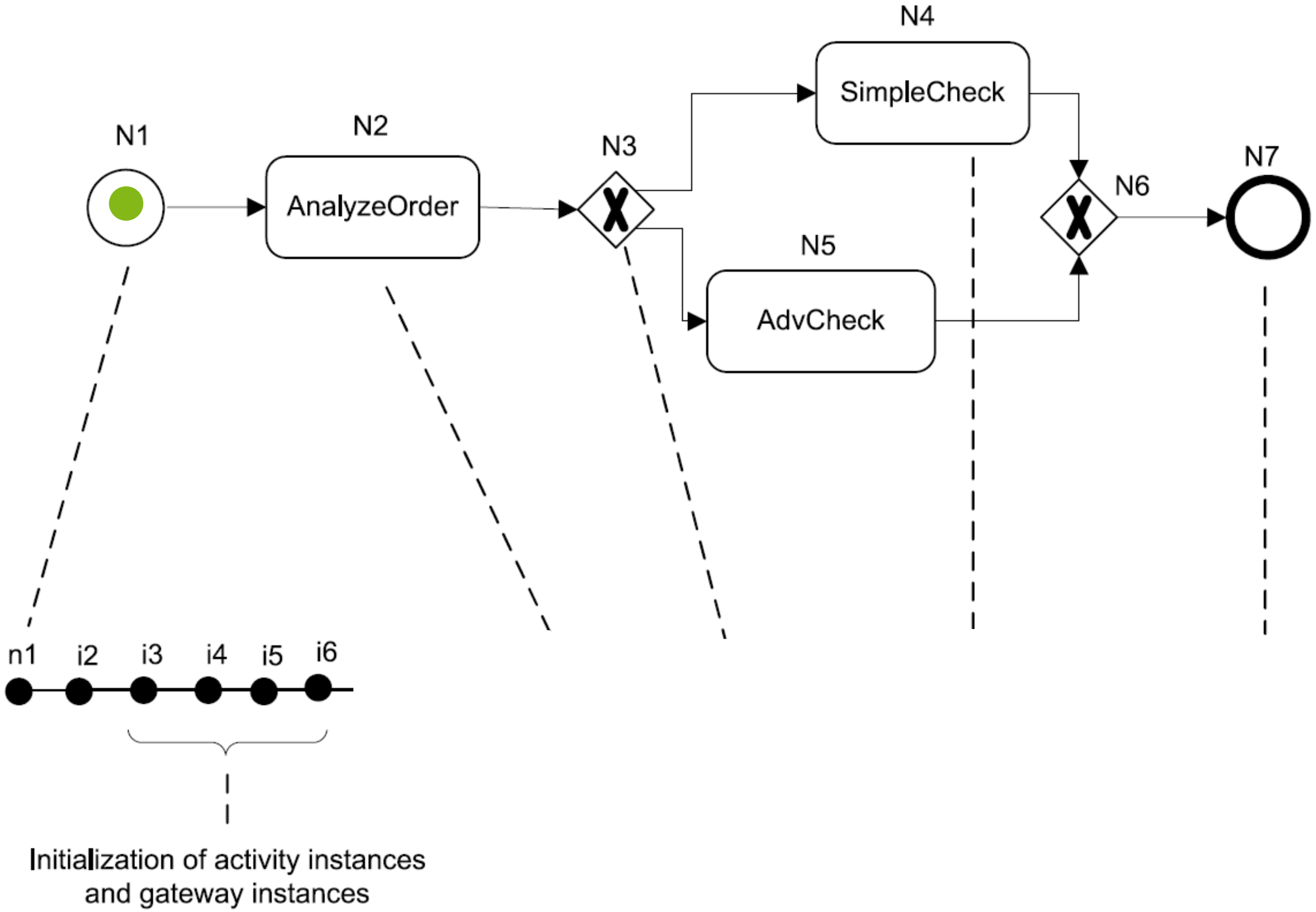
State Changes of a Process Instance



Weske (2019)

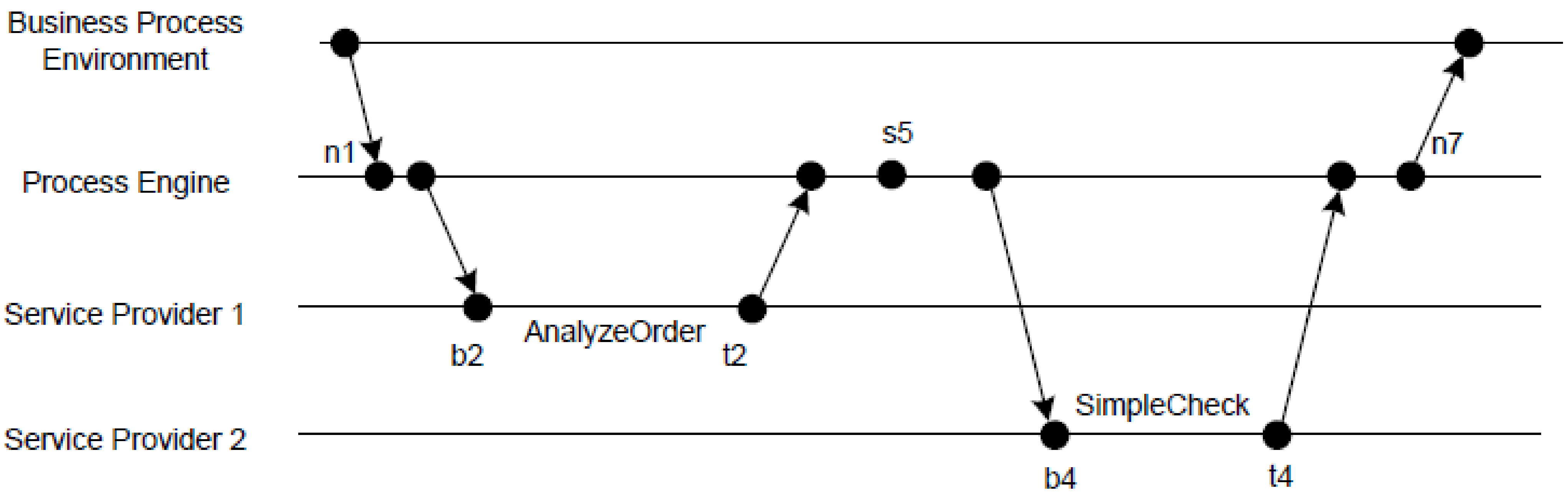
Example for State Changes of a Process Instance

n new event
i initialize
e enable
b begin
t terminate
s skipped



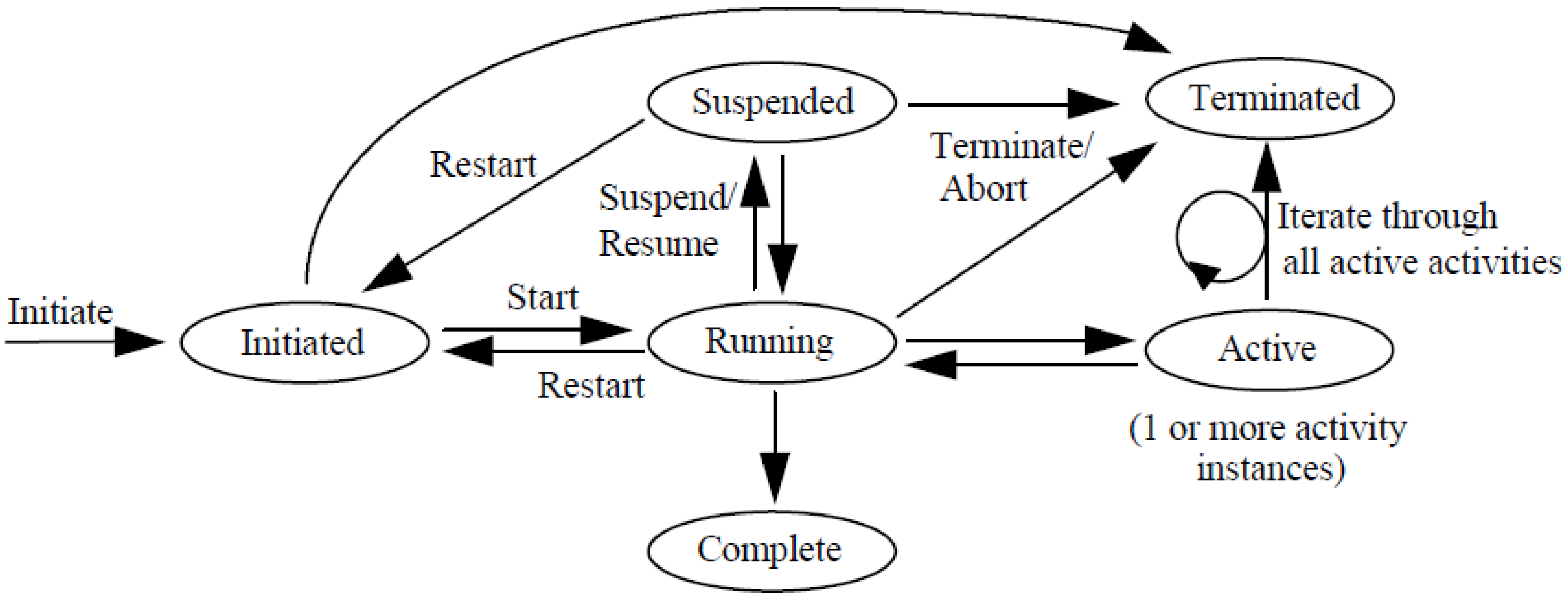
Weske (2019)

Changes to Business Process Execution Environment



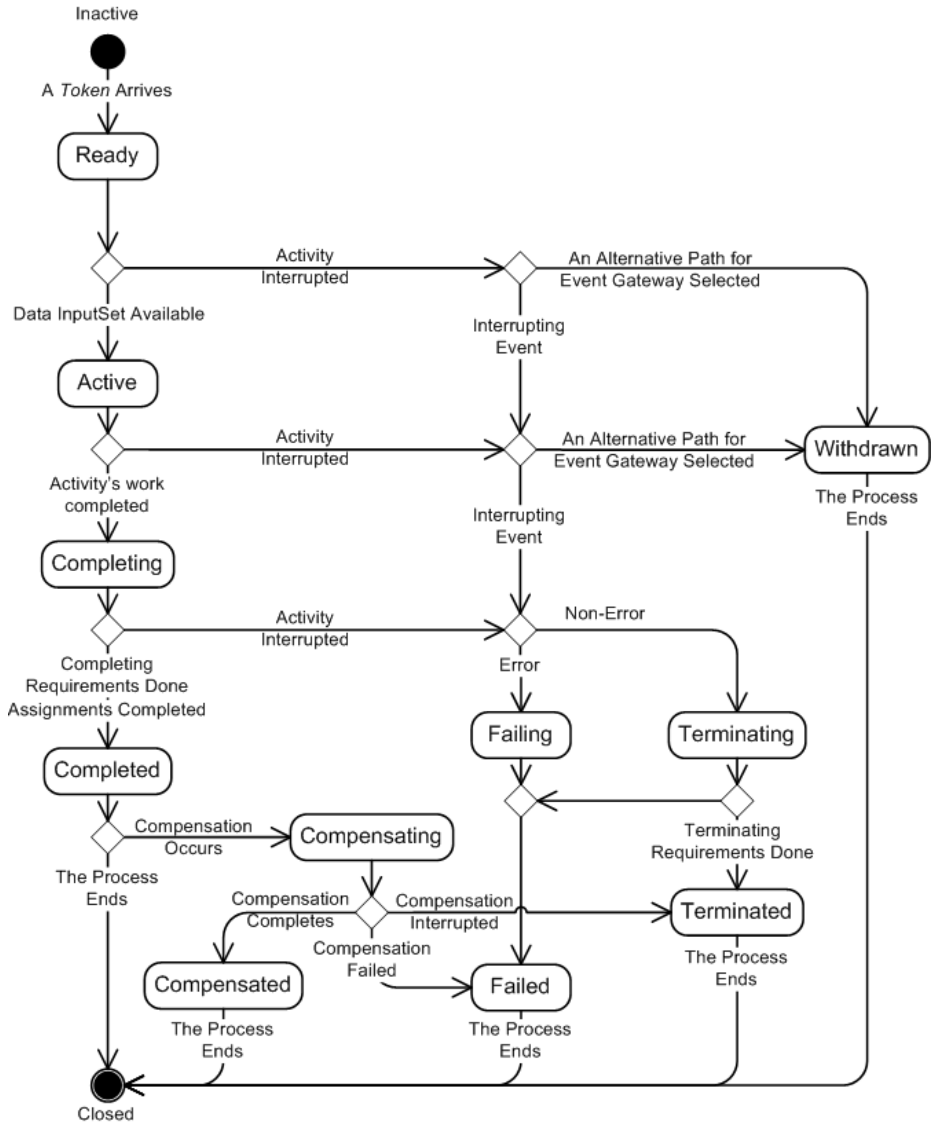
Weske (2019)

States of an Activity



Janiesch (2021), Workflow Management Coaliton (1995)

Lifecycle of an Activity in BPMN



Object Management Group (2011)

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Business Process Management Systems

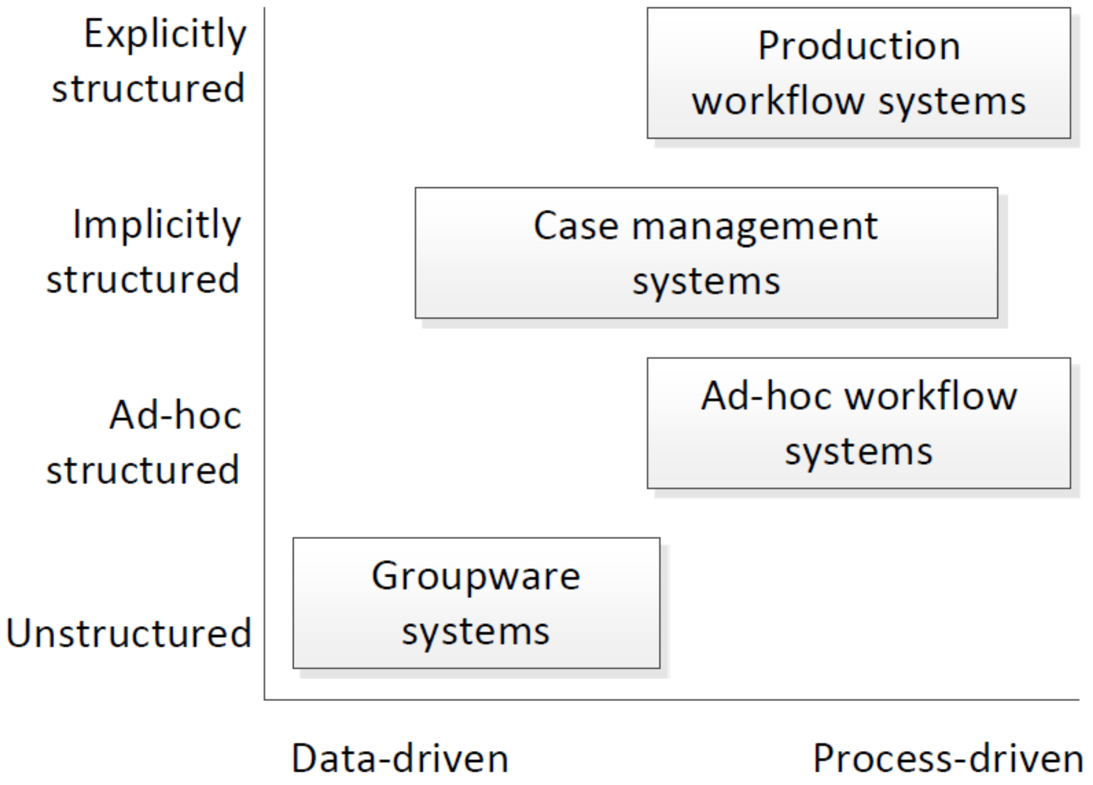


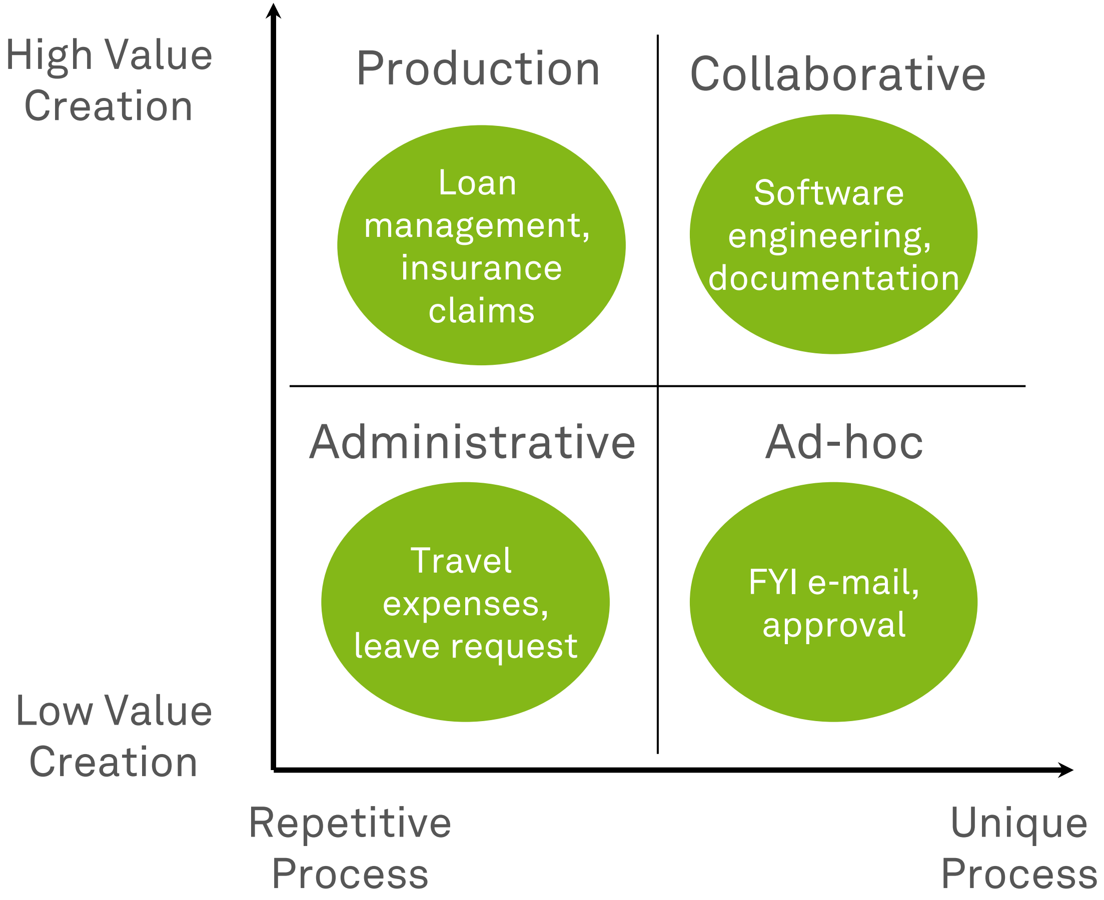

Types of Business Process Management Systems (1)

- **Groupware** systems
  - Do not directly support business processes in a strict sense but allows users to share documents and information
- **Ad hoc workflow systems** systems
  - On-the-fly process definitions that can be created and adapted even during execution
- **Production workflow** systems
  - Most prominent type, distinguished into administrative and transaction processing
  - Work is routed strictly based on process models; data typically handled by complementary DBMS
- (Adaptive) **case management** systems
  - Supports processes that are neither tightly nor completely specified

Dumas et al. (2018)

Types of Business Process Management Systems (2)

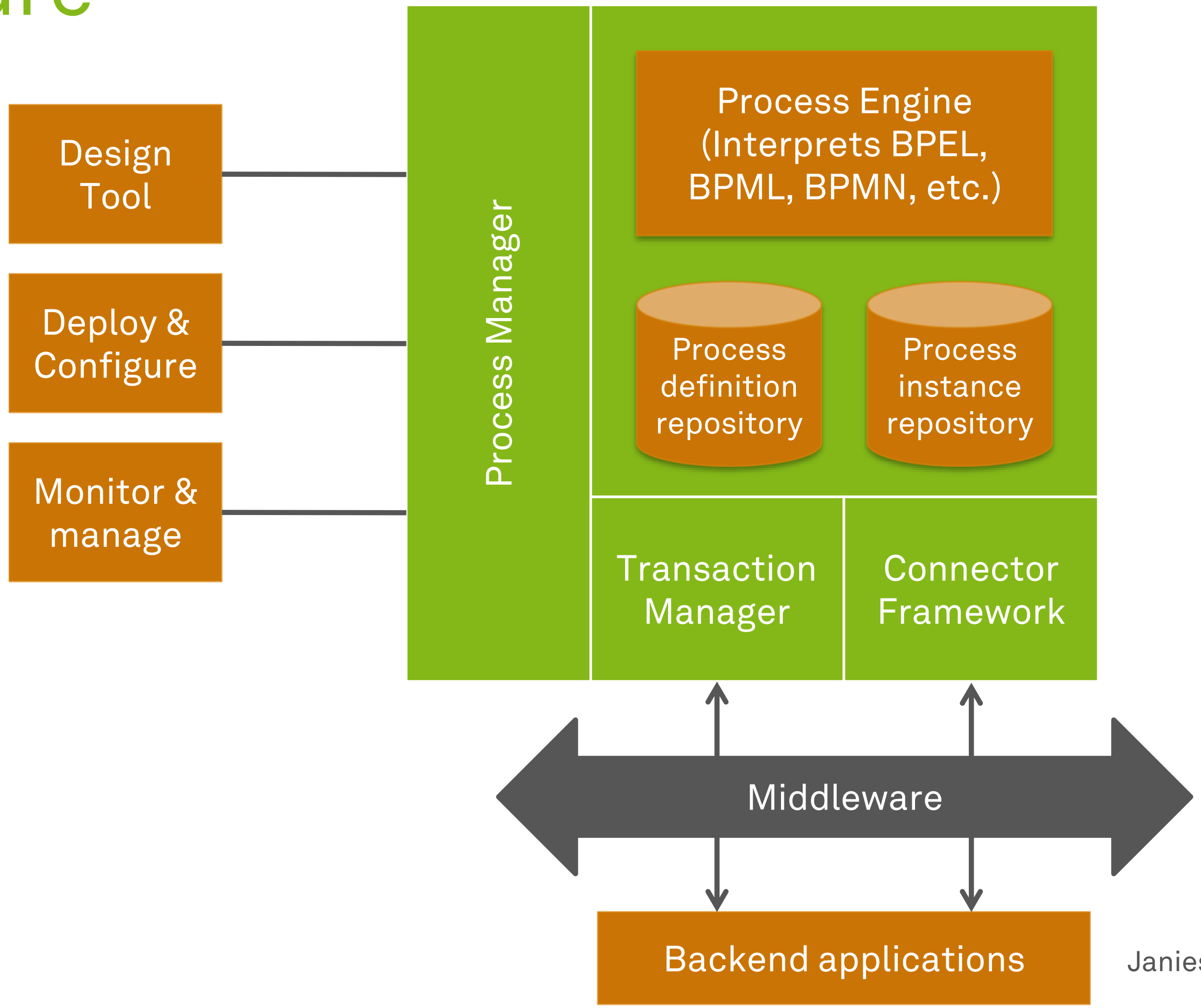




Dumas et al. (2018)

Generic BPMS Architecture

- Provides the **technical platform** for realizing BPM
  - **Process engine**, executing digital process models
  - **Build time** tools for process modeling, simulation and deployment
  - **Run time** facilities for process monitoring and management



Janiesch (2021)

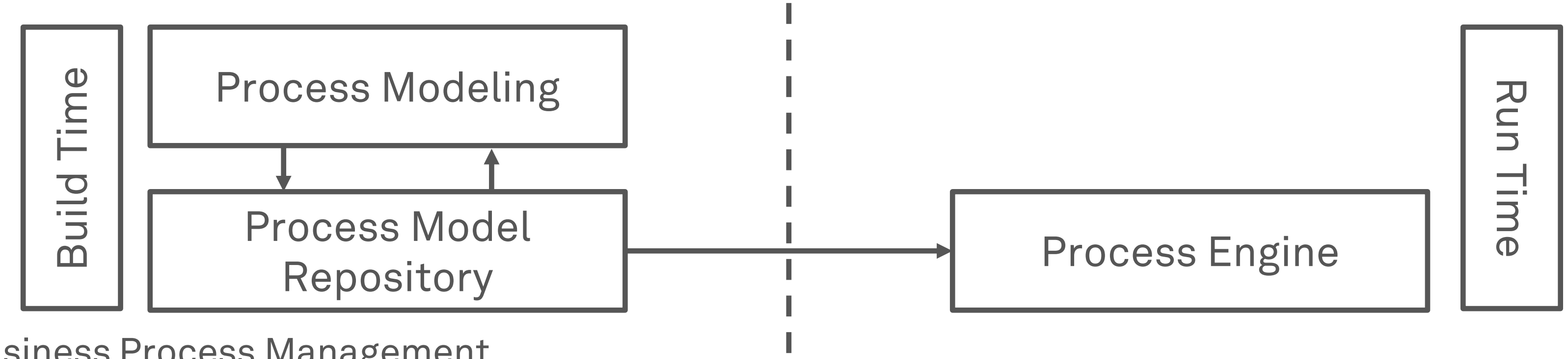
BPMS Architecture

- BPMS standard components
  - **Tooling**
    - Graphically model processes and control BPMS
  - **Repository**
    - Store models in a process definition language
  - **Process Engine**
    - Execute process in a process engine

- BPMS might support additional functionality like
  - Support of different modeling languages
  - Simulation and other analysis methods for process models
  - Support for application integration (EAI middleware features)
  - Monitoring and management of process instances

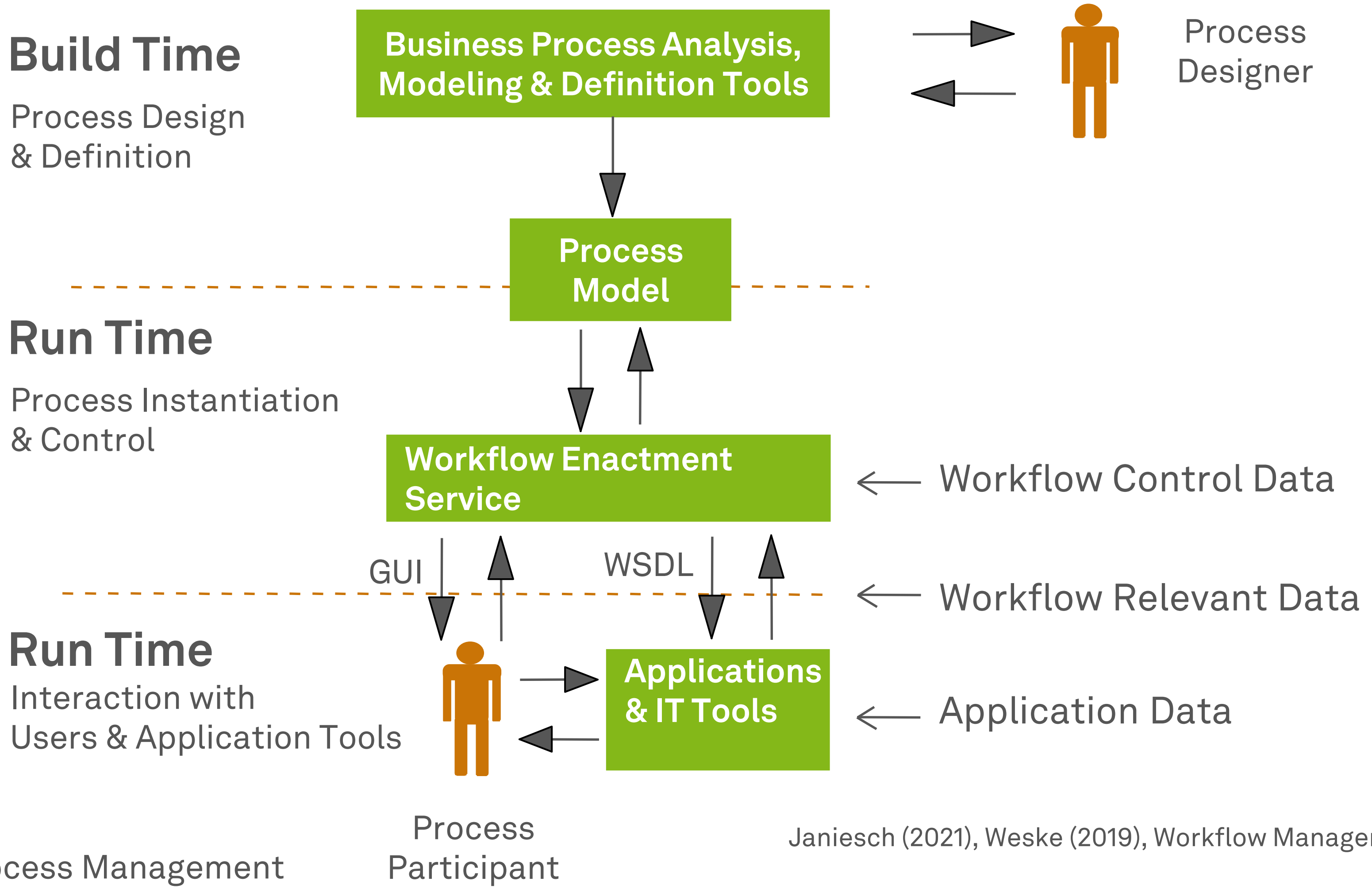
Build Time and Run Time

- A process model has to be **deployed** on the BPMS process model repository in order to be started
- A BPMS creates a **process instance** once the process model is **started**
- A process instance typically **lives** in the BPMS process engine's main **memory**
- The process engine **decides** which activities are executed and **communicates** with client applications
- Traditionally, process instances are **independent of the process model** once being executed



Janiesch (2021), Weske (2019)

BPMS Architecture and Workflow Data



Janiesch (2021), Weske (2019), Workflow Management Coalition (1995)

Advantages of a BPMS

- Workload **Reduction**
  - Straight-through processing
  - Less coordination
  - Less gathering of relevant information
- Flexible System **Integration**
  - Generic functionality of process layer
  - Easier to change process logic
  - Island automation
- Execution **Transparency**
  - Transparency of operational information
  - Transparency of historic information
- Rule **Enforcement**
  - Reducing freedom of executing process
  - Enforce separation of duties
  - Implement control tasks

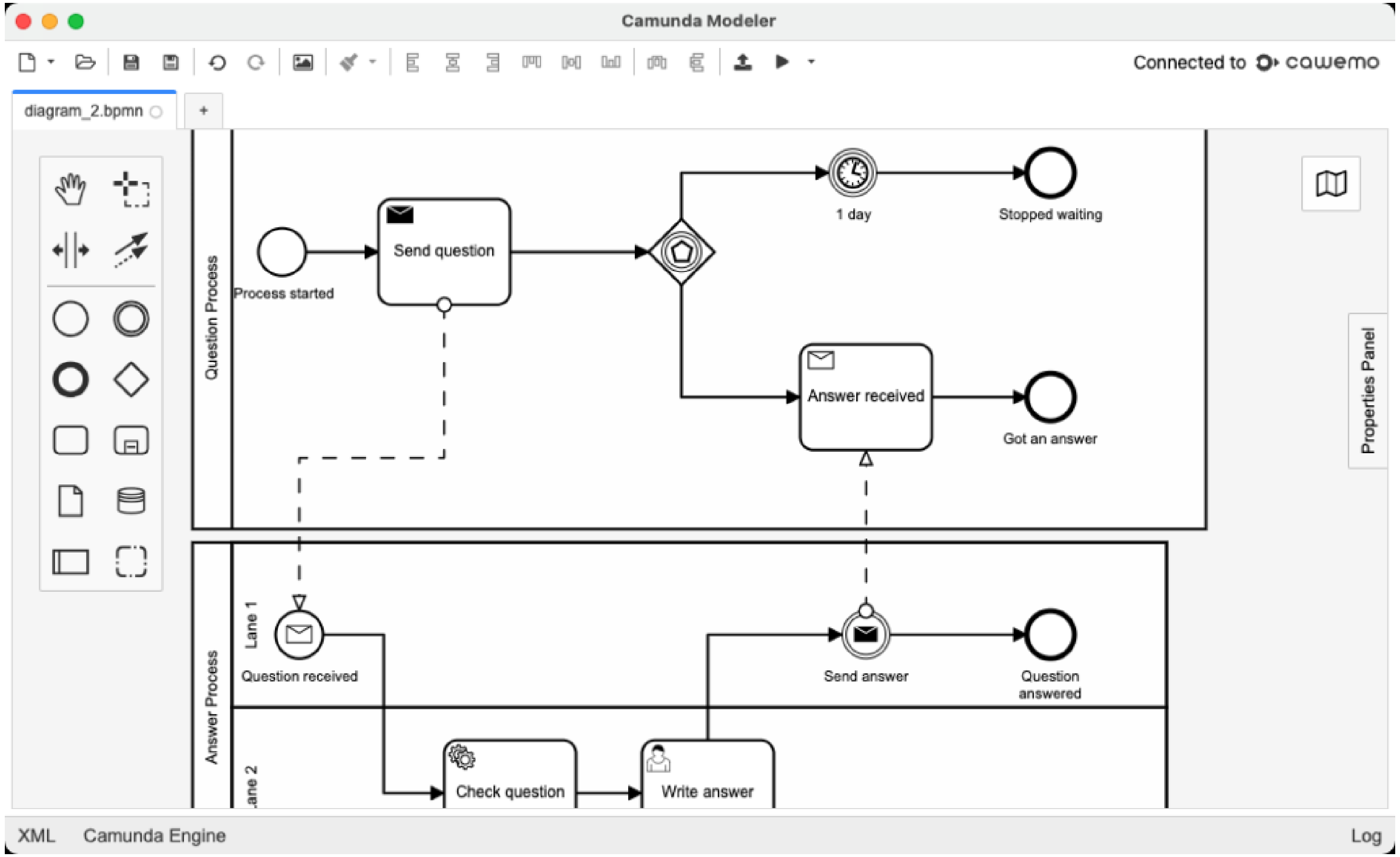
Dumas et al. (2018)

Challenges of BPMS Introduction

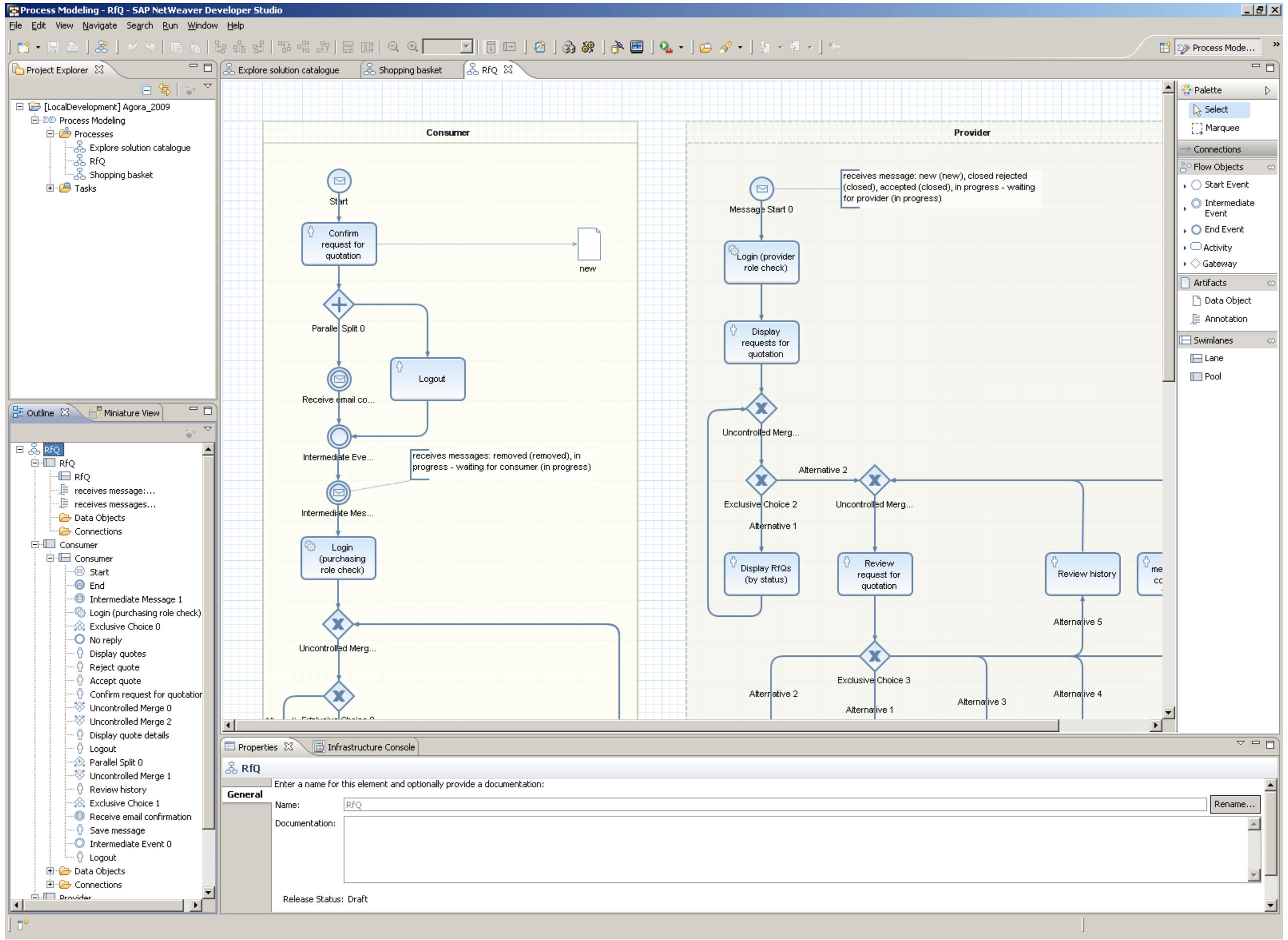
- Technical
  - Applications often not developed from a business process perspective
  - Screen scraping might be required to integrate input and output from **legacy software** (→RPA)
  - **Batch processing systems** do not work with a case concept
  - → Middleware, enterprise application integration (EAI), service-oriented architecture (SOA) and Web service solutions support **integration**
- Organizational
  - **Complexity** due to exceptions
  - Adjust to pace of organizational change
  - Rollout strategy: potential fears of process participants
  - Strong management commitment needed
  - → **Change Management** to mitigate transformation issue

Dumas et al. (2018)

BPMN Tools (1) – Camunda Platform 7



BPMN Tools (2) – SAP NetWeaver BPM



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Robotic Process Automation




Hyperautomation...

- ...is a **business-centric** automation approach...
  - not driven by the IT department
- ...that uses **one or more** technologies...
  - not necessarily enterprise or heavyweight BPM software
- ...to **rapidly automate and augment** business processes...
  - it does not involve change requests, budget evaluations, etc.
- ...in a **scalable** way.
  - elasticity independent of the human factor

hyper-
/ˈhaɪ pər/

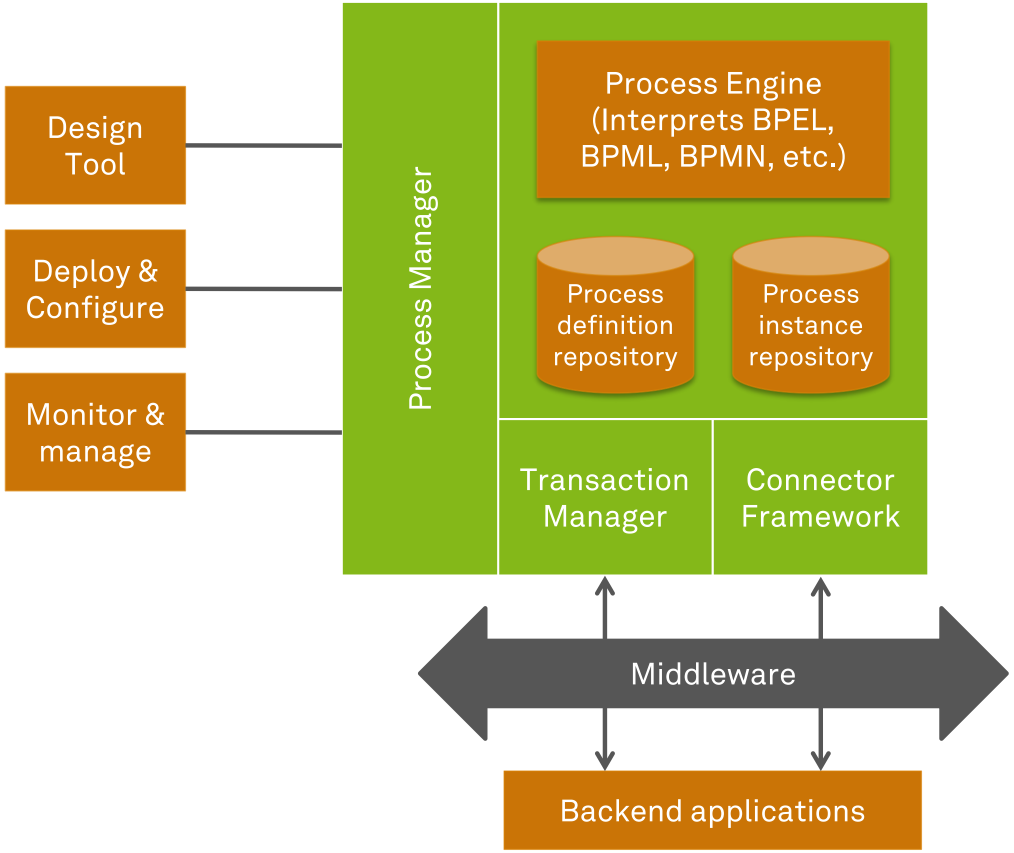
prefix
1. over; beyond; above
2. excess ; exaggeration (hyperbole)

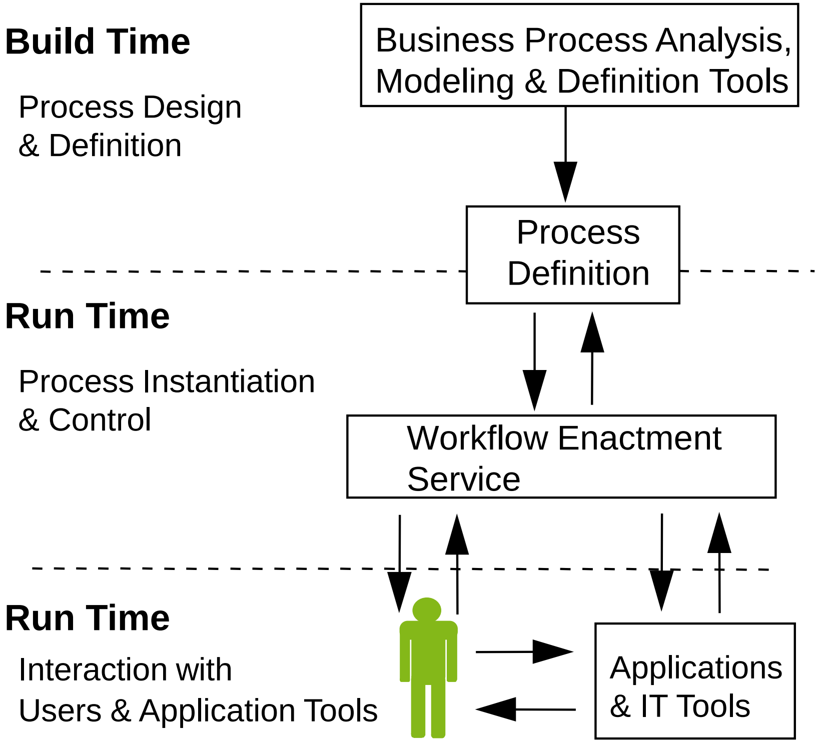
automation
/ˌɔ təˈmeɪ ʃən/

noun
1. the technique, method, or system of operating or controlling a process by highly automatic means, as by electronic devices, reducing human intervention to a minimum
2. ...

dictionary.com (2022), Reijers & Jonker (2020)

Digitalization with Traditional BPMS





Janiesch (2021), Workflow Management Coalition (1995)

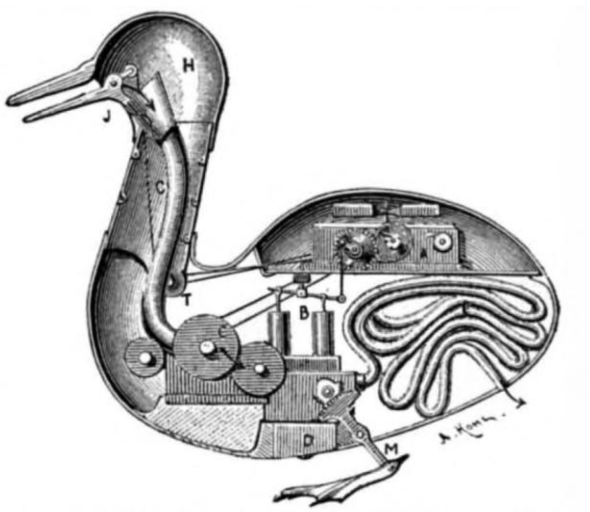
Digitalization with Traditional BPMS

- **BPMS**
  - ...are typically implemented by the **IT department**
  - ...may use multiple technologies to implement activities, but only use **one technology** to rule them all (orchestration)
  - ...**take time** to implement
  - ...do **scale** ☺

Janiesch (2021), Workflow Management Coalition (1995)

Robotic Process Automation

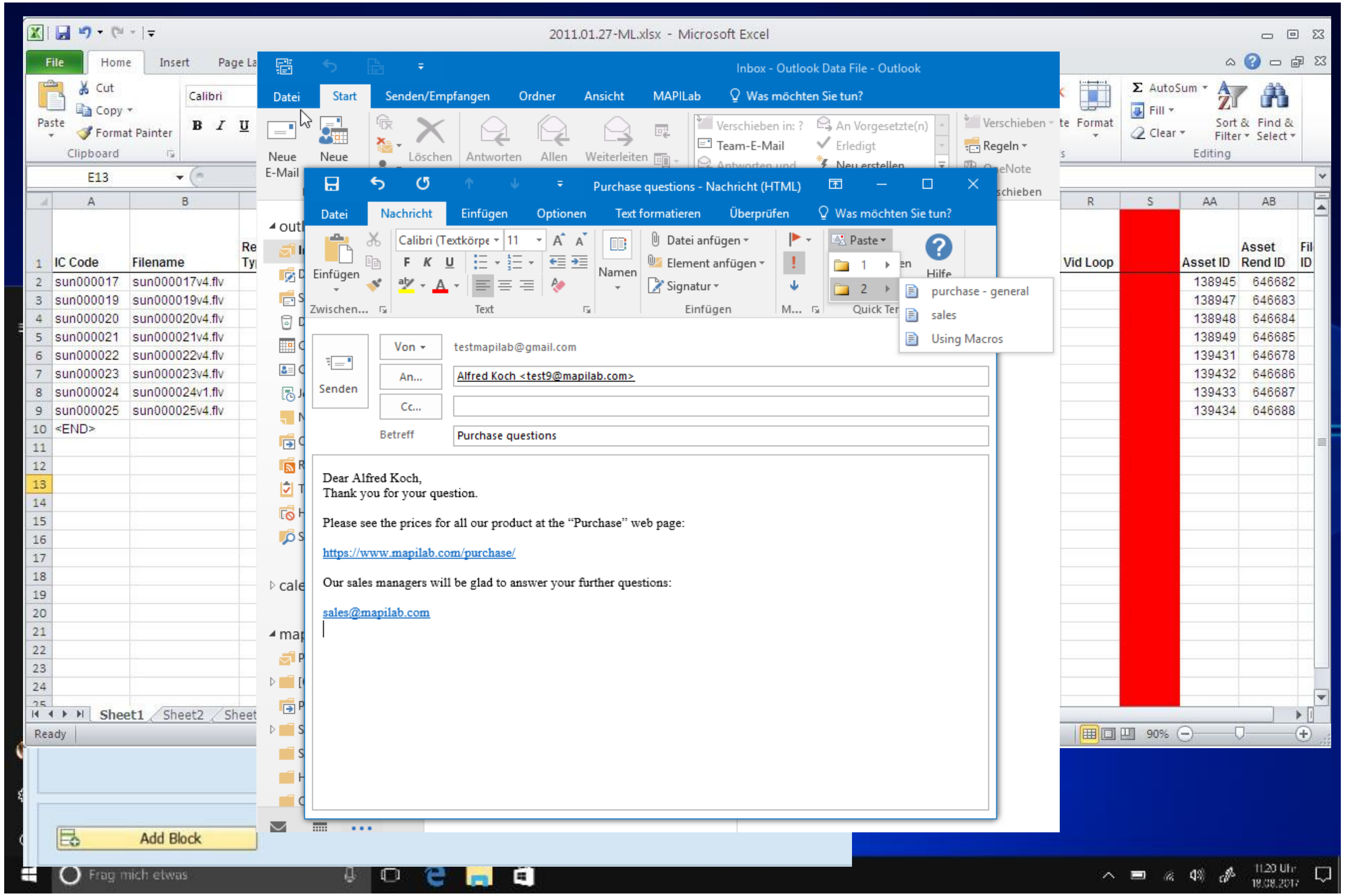
- General Idea
  - Develop an automation (=robot) that is **functionally equivalent** to a human (acting in a specific business process)
- Characteristics of (ro)bots
  - RPA bot is a **single instance** of an RPA system
  - RPA bots are **software** (robots)
  - RPA bots access only the **user interface**
  - RPA bots execute activities using **some rules**
  - RPA is **elastic**
- Result
  - Scalable automation without change in legacy systems



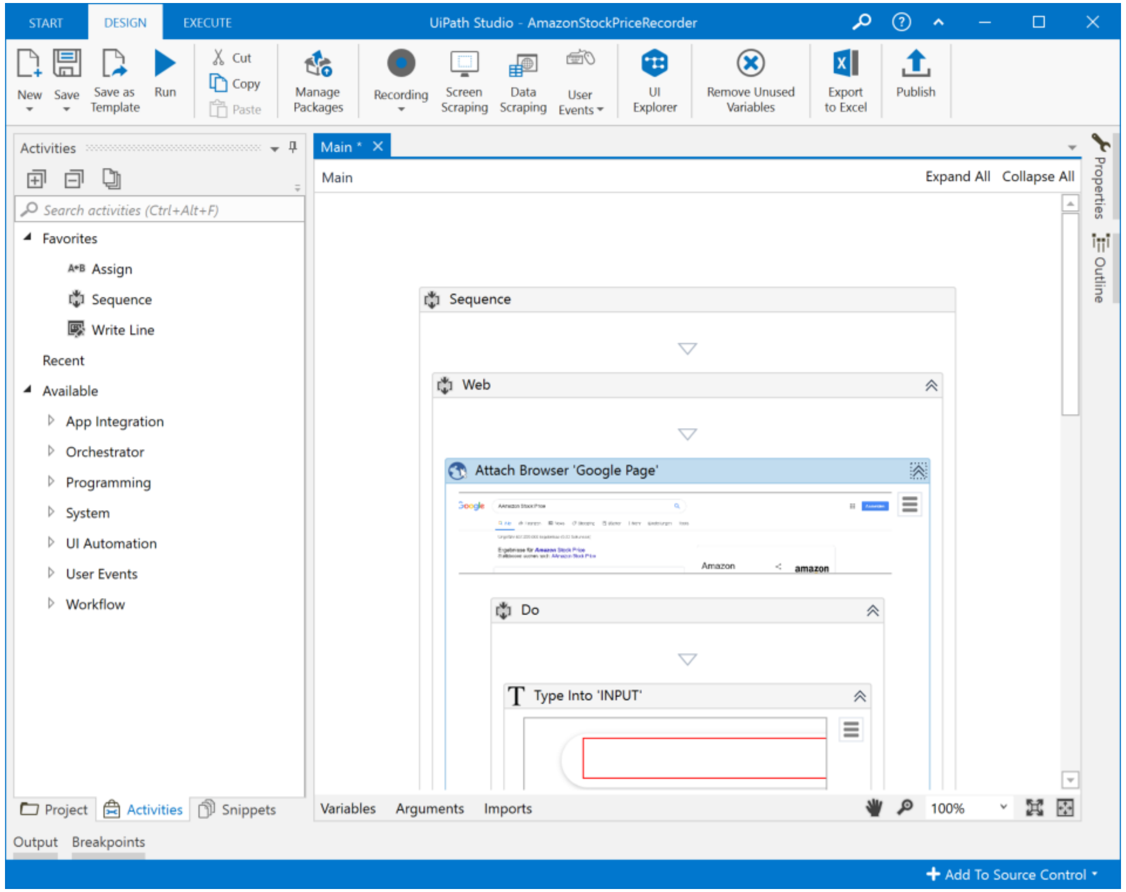
INTERIOR OF VAUCANSON'S AUTOMATIC DUCK.
A, clockwork; *B*, pump; *C*, mill for grinding grain; *F*, intestinal tube; *J*, bill; *H*, head; *M*, feet.

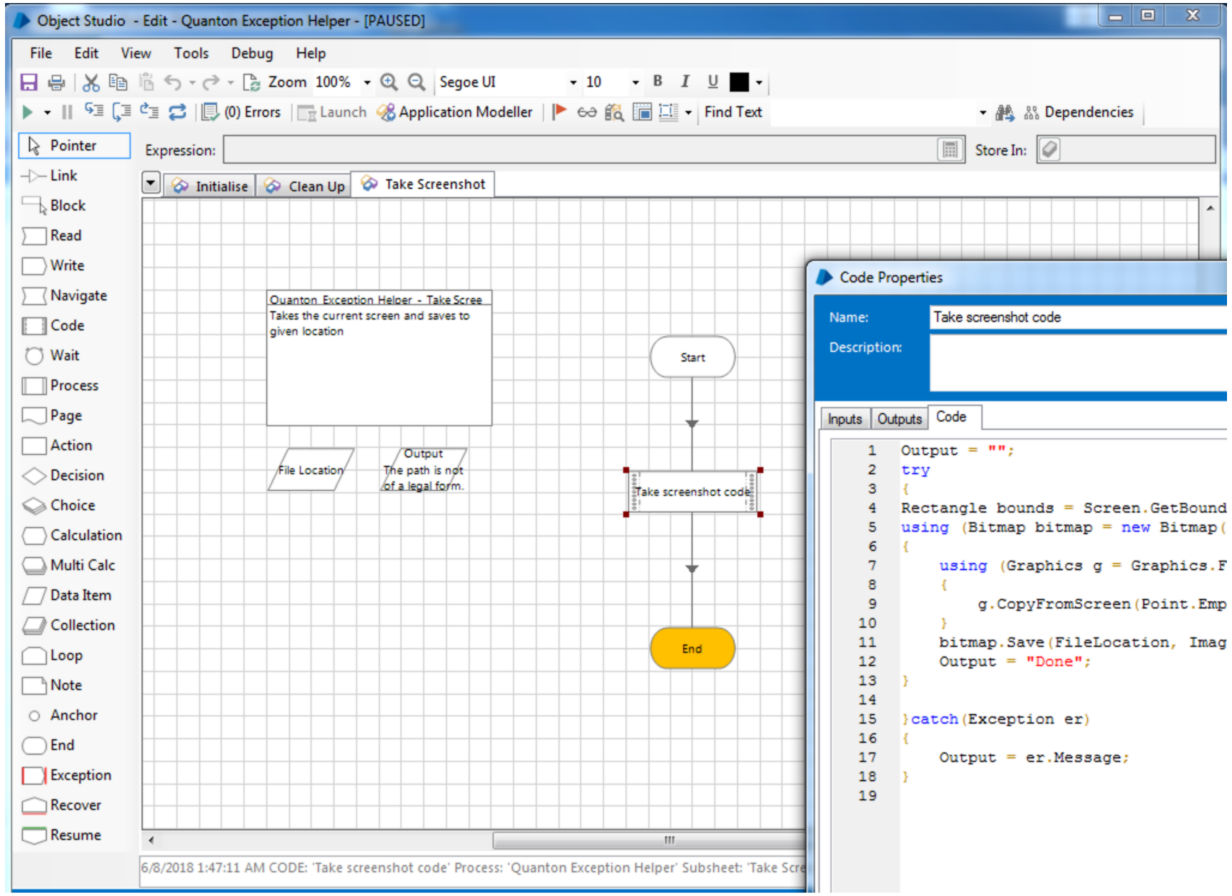
Czarnecki & Fettke (2021), Janiesch (2020), Scientific American (1899)

An RPA Workflow



RPA Software





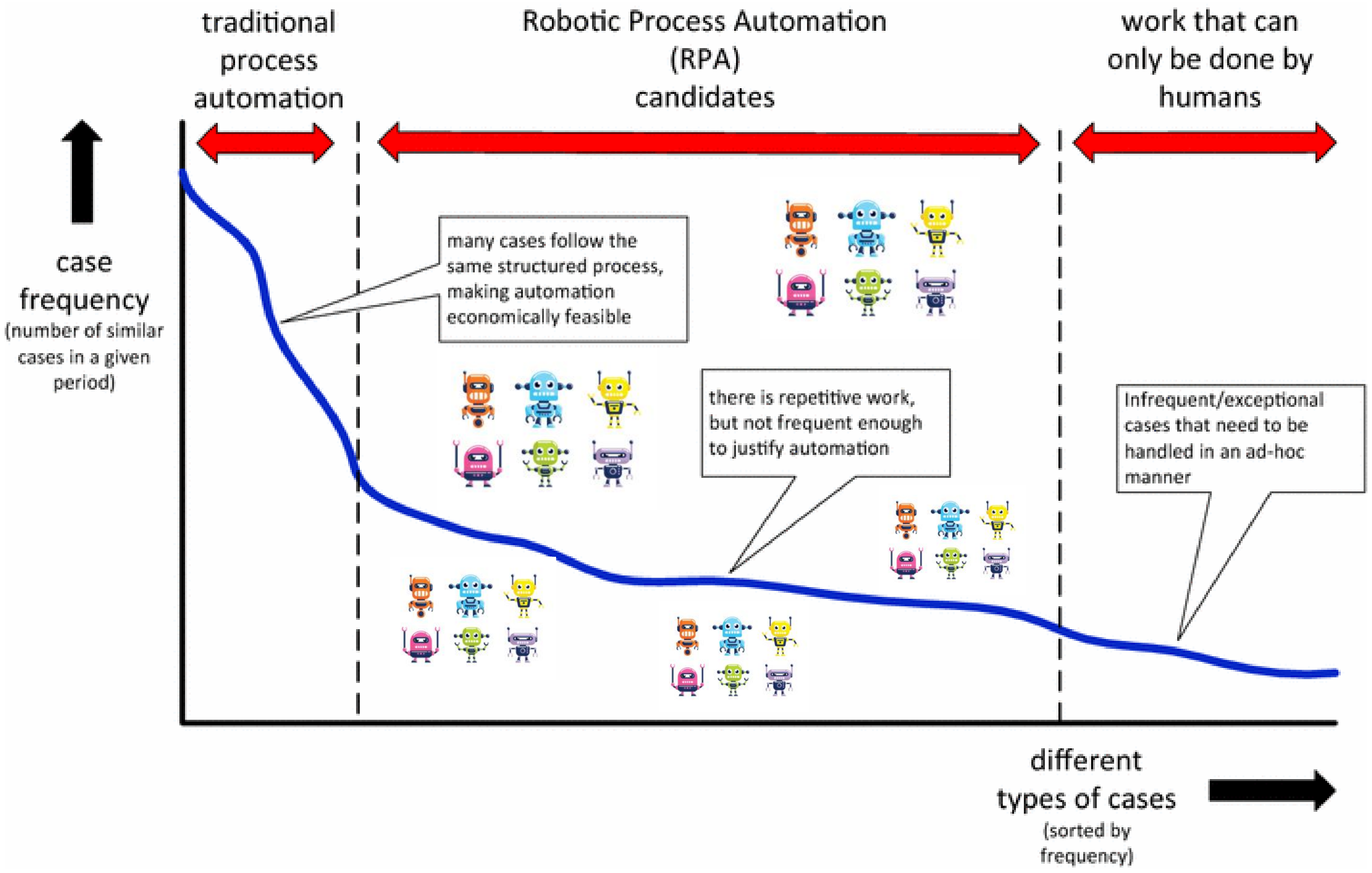
UIPath (2022), Blue Prism (2022)

RPA vs. BPM

RPA	BPM
Digital yet manual tasks	End-to-end processes
Non disruptive	Transformational
Rapid and agile	Long-term
Business driven, DIY	IT-driven

Janiesch (2021), Reijers & Jonker (2020)

BPM and RPA Happily Ever After!



van der Aalst et al. (2018)

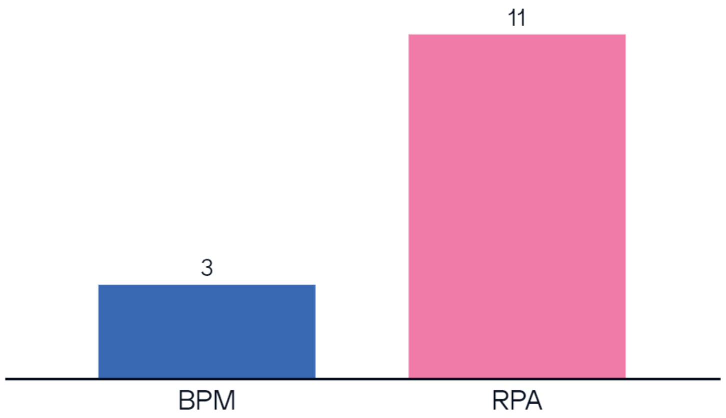
How would you decide? Municipal Health Services (MHS) in the Netherlands during COVID-19

- Covid-19 contact tracing researchers from the MHS complain about the use of system PZ
- People complain that it is illogical and cumbersome
- It has a tricky UI and UX, sometimes crashes and takes an hour to log in again
- The various MHS regions within the system cannot exchange information as they a technically separated, so in formation has to be passed on outside the system
- PZ is MHS's trusted program and cannot be replaced in an instant

- Would you rather have used **BPM** or **RPA** to alleviate the situation?

Reijers & Jonker (2020)

Exemplary Results



fast solution is needed

need a short term solution

More appropriate

A fast and agile approach is needed to ensure functionality

The problems need to be solved within a short period of time

Actually, I would have clicked both.

BPM is to complicated for non IT people to understand and implement in a short time

Reijers & Jonker (2020)

Learning Goals of Today

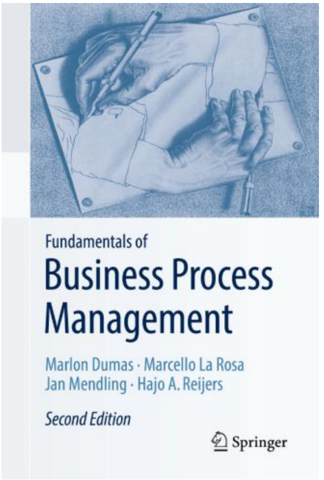
- Understand the evolution of process technology
- Understand workflow execution semantics
- Understand BPMS
- Understand RPA

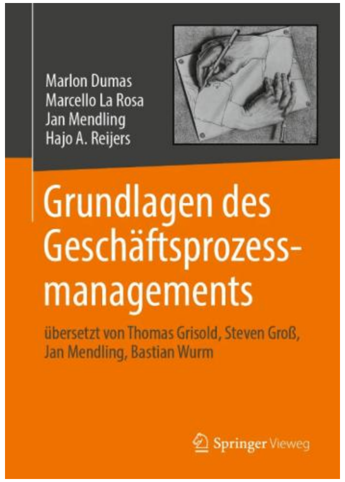
Recap Questions

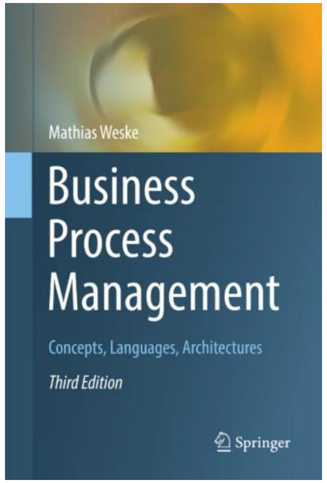
- What are execution semantics and how does a BPM system step through a process?
- What are the principal components of a generic BPM system and how are they arranged?
- What is the relation of robotic process automation to BPM systems?

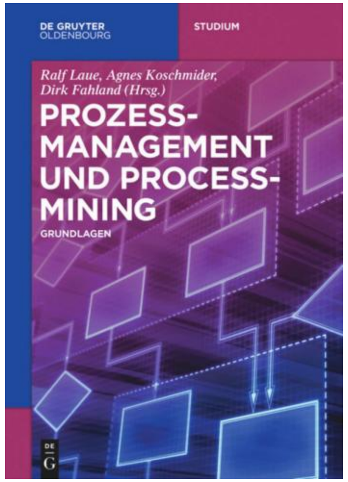
Literature

- **Section 9:** Process-Aware Information Systems
- **Section 2:** Evolution of Enterprise Systems Architectures
- **Section 3:** Business Process Modelling Foundation
- **Section 8:** Business Process Management Architectures
- **Section 9:** Geschäftsprozessmanagement-systeme und Robotic Process Automation









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Any Questions about Process Technology?

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